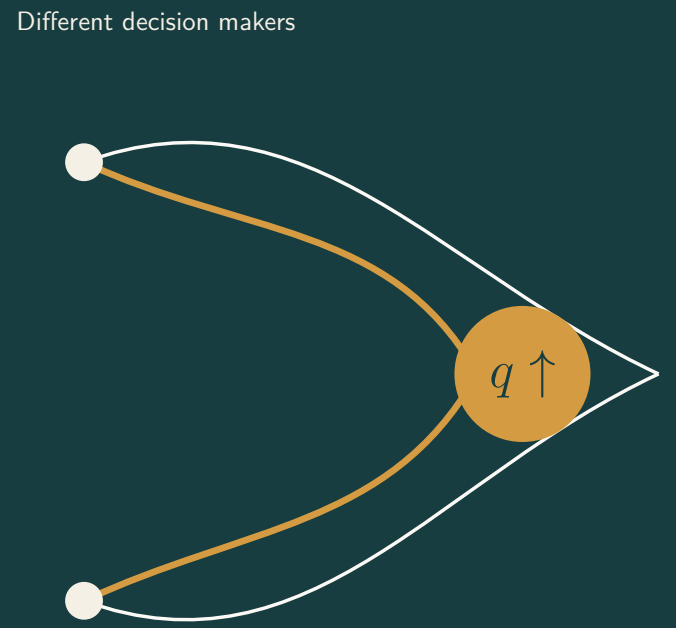
When everyone follows the same best advice.

Better predictions can concentrate choices.

When opportunities are scarce, accuracy and economic value can pull apart.

From Braess's paradox to shared advice

Mathematical model · Synthetic computations · Not peer reviewed



Same destination.
More competition.

From roads to recommendations.

1968 / 2005 BRAESS

A new route can worsen traffic equilibrium.

2018 ACEMOGLU ET AL.

More route information can hurt travelers.

2021 KLEINBERG & RAGHAVAN

Common rankings can reduce collective hiring performance.

2025 / 2026 BAEK ET AL.

Congestion-aware differentiation matters.

THE MODEL'S ACCOUNTING IDENTITY

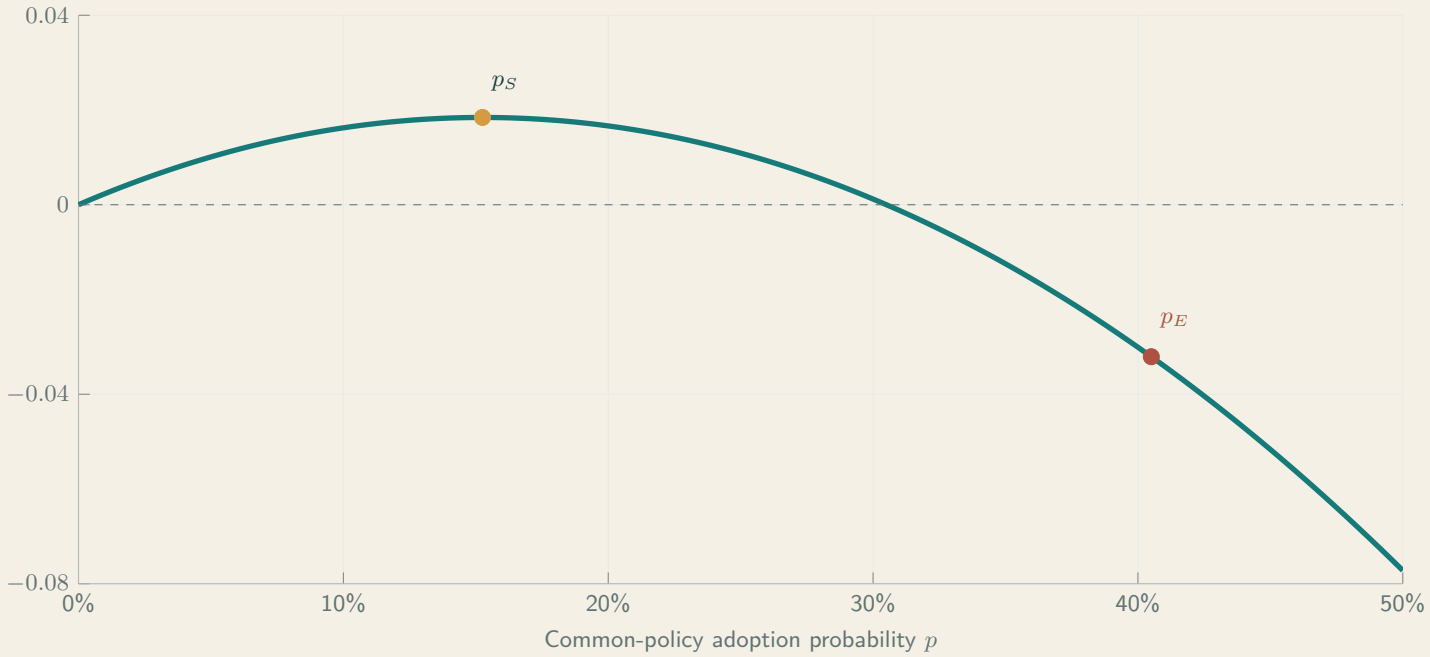
$$\frac{\mathbb{E}[W]}{N} = \mu - c(N-1)q$$

μ: expected opportunity value
q: probability two choices coincide
c: resource cost per competitor

Sources: manuscript §2 and Proposition 3.1; full references in the accompanying source.

Individually attractive. Collectively costly.

Exact payoff change from private-only baseline; adoption range shown: 0–50%.



15.3%

Payoff-maximizing independent adoption

40.5%

Equilibrium adoption
Loss: 0.03208 per person

$N = 100,\ a = .6,\ b = .8,\ c = .02,\ d = 2.$
Arbitrary payoff units.

The loss can survive rational adaptation.

Each person sees only their own private and shared recommendations.
At the specified interior equilibrium, the policy outcome remains implementable.

THE SIGNALS AGREE

Follow the recommendation.

Conditional gain: $401/875 > 0$

THE SIGNALS DISAGREE

Either action is optimal.

Mixing with $p_E = 401/990$ sustains equilibrium.

Proposition 6.1: an existence result under a specific information structure. The figures above use the illustrative calibration.
The model does not establish real-world economic losses, dynamic chaos, or uniqueness.

github.com/triggeredcode/shared-intelligence-competing-decisions · AI was used.

A rethinking of established research.

This note reworks familiar congestion and information externalities in a small model of shared advice. Its original research contribution has not been established.

FOUNDATIONS AND CLOSE PRECEDENTS

Braess (1968; English translation 2005): traffic planning.

Acemoglu et al. (2018): informational Braess' paradox.

Kleinberg & Raghavan (2021): algorithmic monoculture.
Baek, Bastani & Chen (2025/2026): strategic hiring.

FURTHER COMPARISON REQUIRED

Kleinberg, Sinanaj & Tardos (2026): price of anarchy of algorithmic monoculture.

Chi & Yang (2026): shared intelligence and firm entry.
Chiang et al. (2026): recommendation-induced congestion.

The explicit formulas and Bayesian implementation are results within this model; their novelty relative to prior work remains unverified. Mathematical checks establish internal consistency, not originality.

Full titles, links, additional references, and limits of the comparison: RESEARCH_CONTEXT.md in the repository.